

APF-UF Series

Ultra-thin Force Sensor



Description

Angst+Pfister's newest technology is an ultra-thin force sensor with a thickness of **under 1mm**. This compact design allows integration into even the most space-constrained applications. With the encasing of a durable elastomer, the sensor is fully protected from moisture and environmental influences, ensuring reliable performance in diverse conditions.

The sensor measures applied force under compression and is suitable for a wide range of applications.

Geometry, mounting style, connector type, force range, and digital interface can be customized to project requirements.

Mechanical Properties¹

Diameter of sensing element	15	mm
Height of sensing element	<1	mm
Height with elastomer and PCB components	3.6	mm
Total diameter	22	mm

¹ Values for this specific design

Electrical Properties²

Digital interface	UART	Protocol
Output data rate	20 Hz	Hz
IO voltage	3.3	V
Supply voltage	2.3-5.5	V

Operating Conditions

Reference temperature	25	°C
Operating temperature range	0...120	°C
Compensated temperature range	0...50	°C
Humidity range (non-condensing)	0...95	%rH

Measurement Characteristics (3.3V, 25°C)

Measuring range (Full scale output – FS)	0 to 10N	N (range)
Repeatability ³	± 0.5 reading + 0.05% FS	
Accuracy (25°C) ⁴	± 1% reading + 0.05% FS	
Resolution	5	mN
Overload	100	N
Rise time (t ₉₀) ⁵	<50	ms

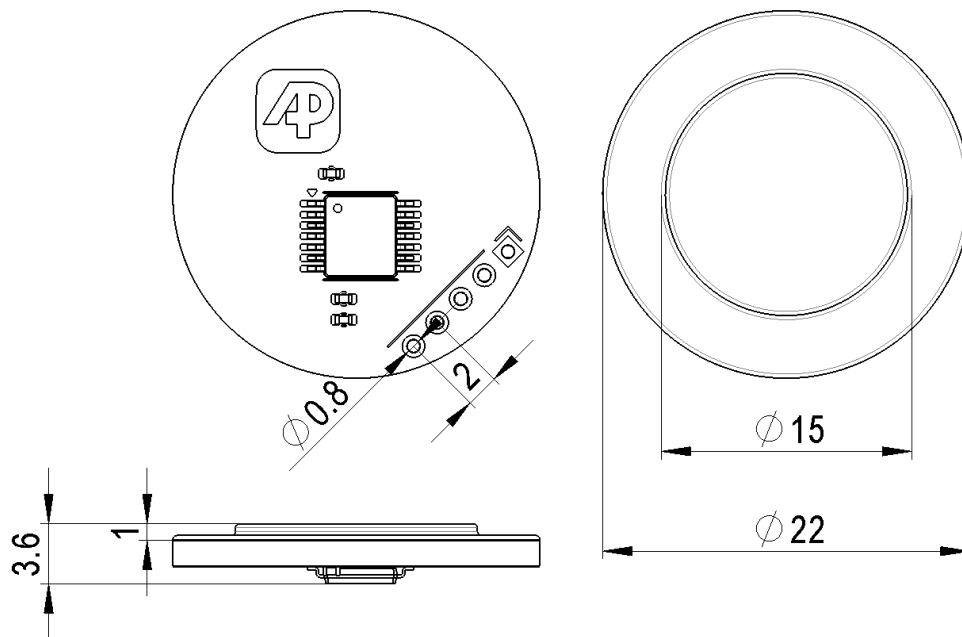
² Properties refer to the microcontroller attached via a cable to the sensing pad. For larger sensor designs, the microcontroller can be mounted directly on the sensing pad PCB.

³ Minimum and maximum deviation of the output signal after 50 cycles at 10 N includes repeatability errors from force introduction misalignments and application speed of the test setup.

⁴ Minimum and maximum deviation of the output signal at peak force over $n = 50$ loading and unloading cycles tested at 200 mN, 1 N, 5 N, and 10 N includes static drift, hysteresis, non-linearity, and repeatability of force application. Hysteresis and static drift are compensated using DSP.

⁵ Step wave applied. Measurements limited by output data rate.

Dimensions



Application Example – Thin Kitchen Scale

A statically defined three-point configuration ensures even load distribution and stable measurements. It is suitable for applications such as liquid tank weighing, kitchen scales or general load sensing. Additionally, the three sensor signals enable calculation of the weight distribution and the system's centre of mass.

Measuring range (Full scale output – FS)	0 to 3	kg (range)
Repeatability ⁶	± 0.5 reading + 0.05% FS	
Repeatability with 100g	0.03	% FS
Accuracy ⁷	± 1% reading + 0.05% FS	
Sensitivity [0 – 3kg]	0.5	counts /g
Resolution	1	g
Max overload	20	kg

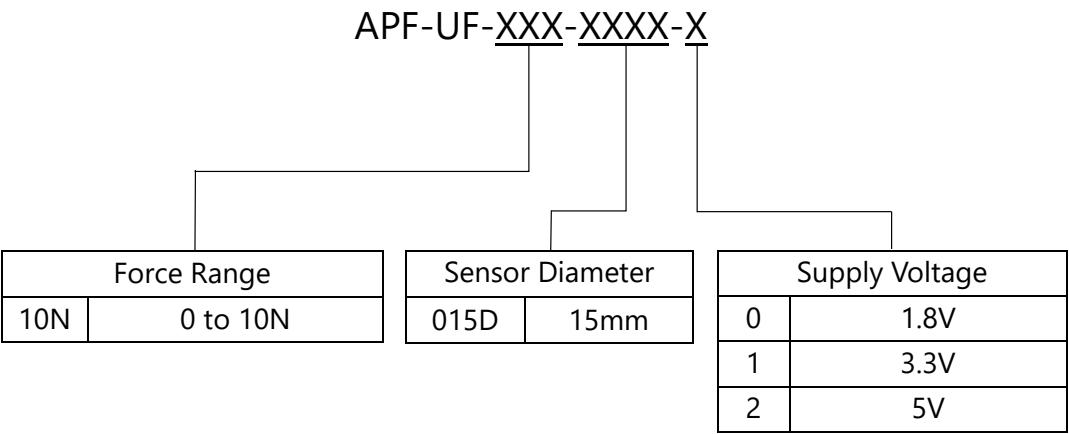
Additional Information

The sensors must be connected to a microcontroller that converts the raw I²C signal into a UART signal and outputs the compensated and calibrated force data. The accuracy can be further improved by adjusting the calibration parameters for the specific application. The microcontroller PCB can be used for up to four sensors.

⁶ Minimum and maximum deviation difference of the output signal after 50 cycles with 1000g.

⁷ Minimum and maximum deviation difference after n = 50 cycles. Tested for 100g, 500g, 1000g. Including static drift, hysteresis, force non-linearity and repeatability of force application. Hysteresis and static drift are compensated using DSP on the microcontroller.

Ordering Information



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